

Qatar-10b

R-0010

Benchmark run · workflow W8-benchmark · record 2026-bench_Qatar-10_260701 · created 2026-07-08T03:37:09+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461222.648 +/- 0.015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.051
Transit depth (Rp/Rs)^2	3.0 +/- 1.77 [%]
Orbital Inclination (inc)	88.27 +/- 2.7
Airmass coefficient 1 (a1)	34.23 +/- 21.48
Airmass coefficient 2 (a2)	-3.0 +/- 1.21
Scatter in the residuals of the lightcurve fit is	86.69 %
Best Comparison Star	#1 - [236, 299]
Optimal Aperture	2.45
Optimal Annulus	6.0
Transit Duration (day)	0.059 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.173 ± 0.051 vs 0.1265 (deviation 0.91σ)
Duration fit vs published	0.059 d vs 0.1154 d (deviation 3.53σ)
Mid-time O–C	-14.0 min
Depth SNR	3.4
Residual scatter	86.69 %

Light curve

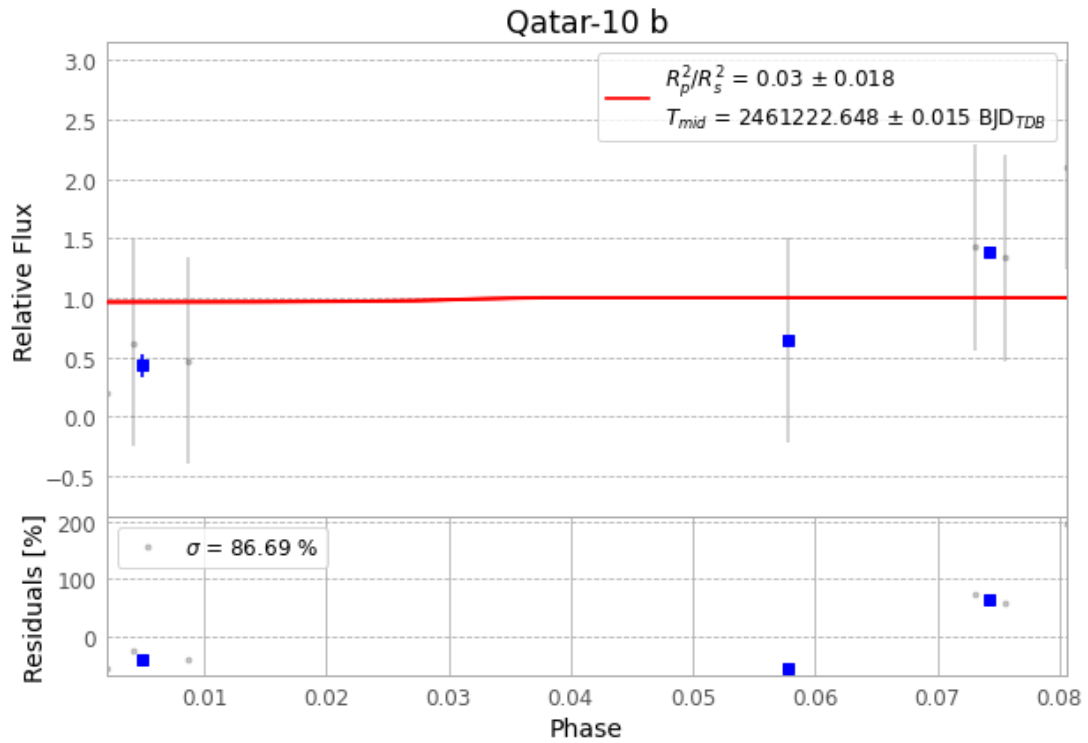


Figure 1 — FinalLightCurve_Qatar-10 b_2026-06-30.png (SHA-256 52e61206ca5e5ba9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [304, 267], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5a9fe0fc2a5cfa66...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

78 FITS frames (51 MB), Qatar-10260701030117.FITS → Qatar-10260701065115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-10-260701.log (3f63fa685a4f...), FinalLightCurve_Qatar-10 b_2026-06-30.png (52e61206ca5e...), FinalLightCurve_Qatar-10 b_2026-06-30.csv (b3b4da5d01c3...)

Compute resources

Wall time 176.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.